Functions And Graphs

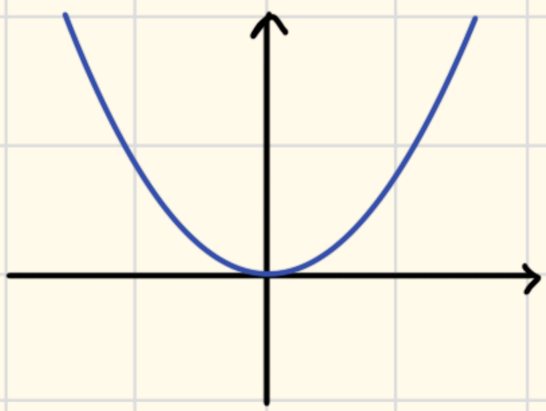
There are many functions that model different problems.

Function Name	Parent Function
∘ Straight - line	$y = x$
∘ Quadratic	$y = x^2$
∘ Cubic	$y = x^3$
∘ Reciprocal	$y = \frac{1}{x}$ OR $\frac{1}{x^2}$
∘ Exponential	$y = 2^x, 3^x, \ldots$
∘ Circle	$x^2 + y^2 = r^2$
∘ Trigonometric	$y = \sin\theta, \cos\theta$ OR $\tan\theta$

Functions And Graphs



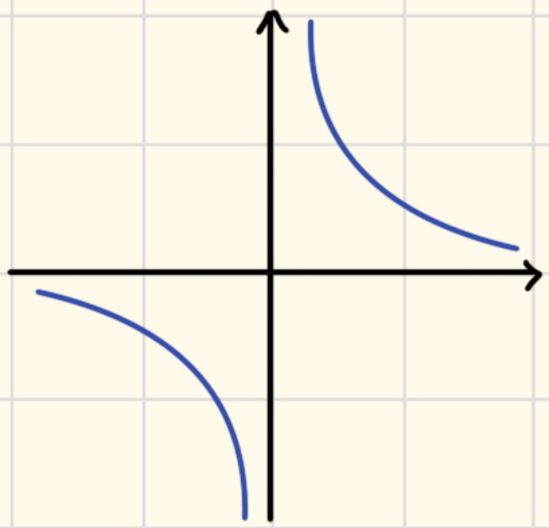
Straight Line



Quadratic



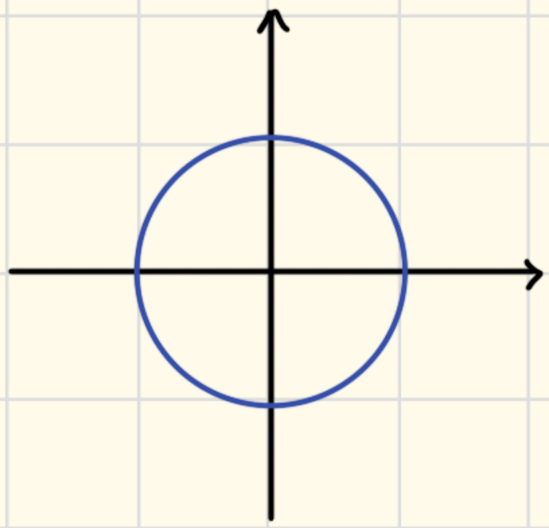
Cubic



Reciprocal



Exponential



Circle

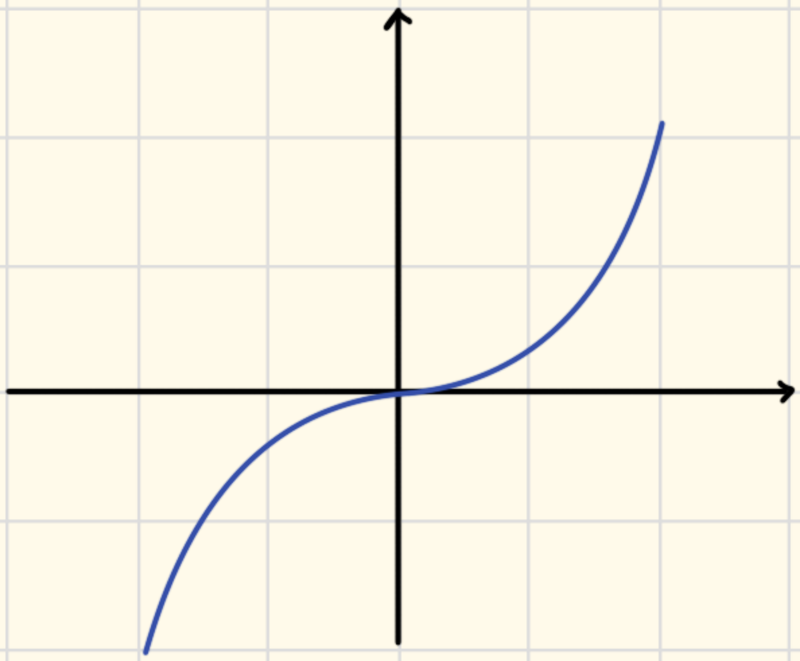
CUBIC

Cubics are functions of the form

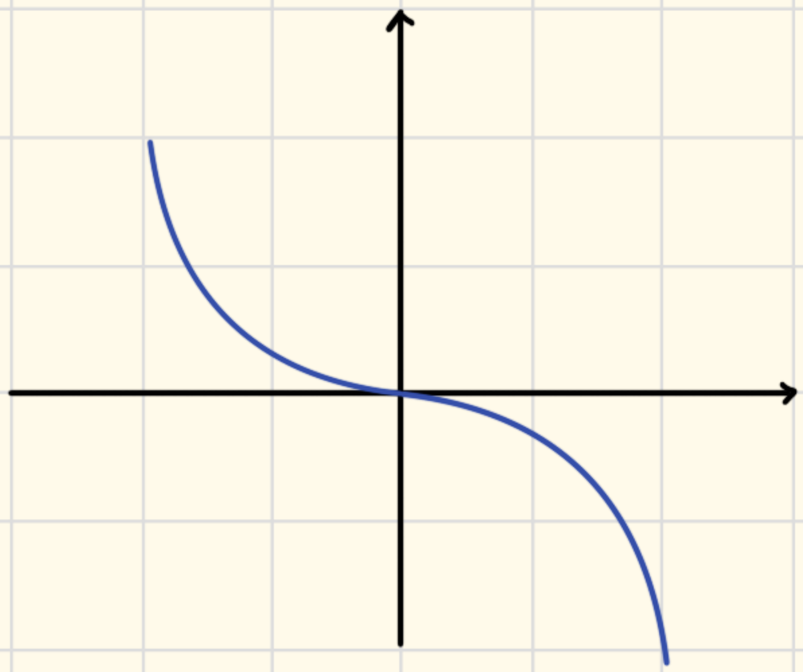
$$y = ax^3 + bx^2 + cx + d$$

degree 3!

$y = x^3$



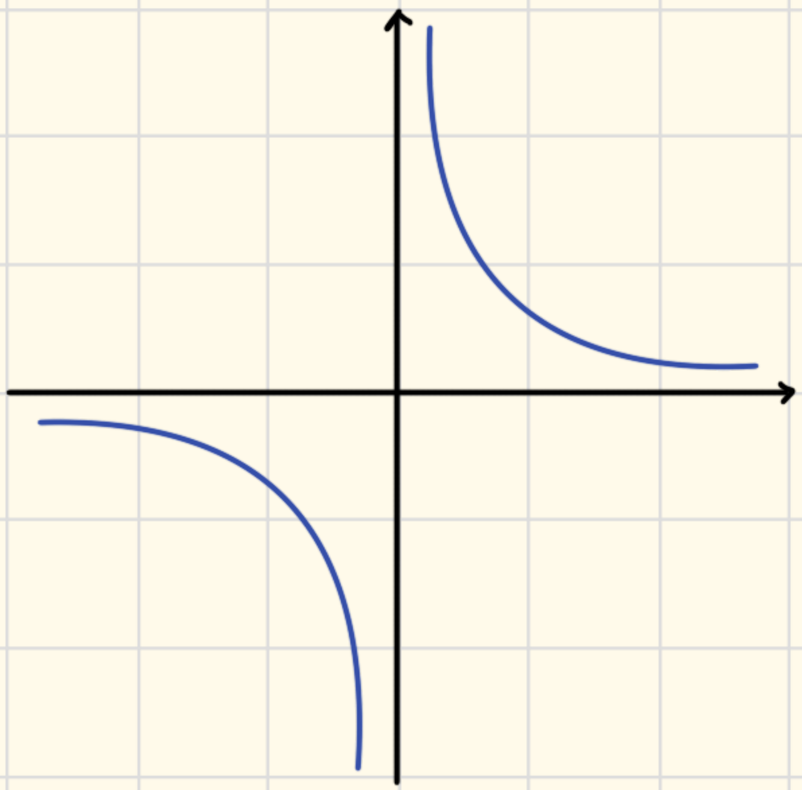
$y = -x^3$



RECIPROCAL

Reciprocals model inverse relations.

$$y = \frac{1}{x}$$



$$y = -\frac{1}{x}$$

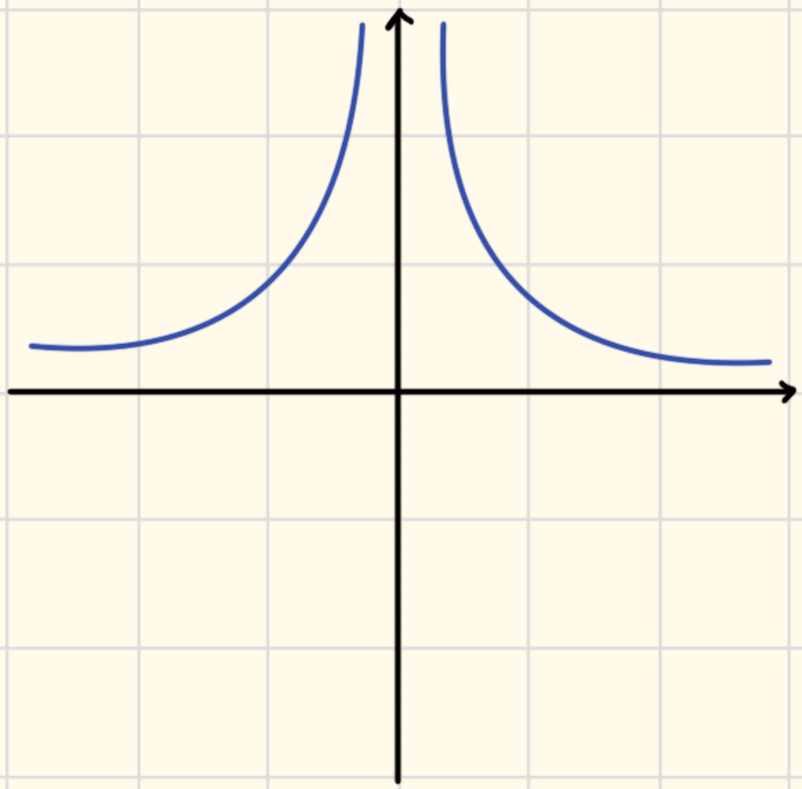


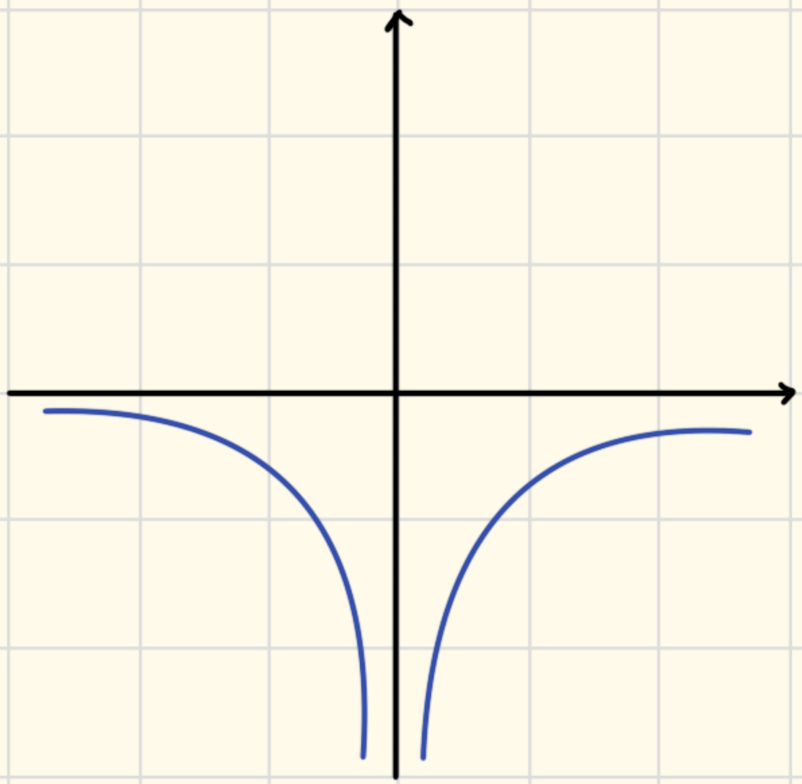
RECIPROCAL

Reciprocals model inverse relations.

$$y = \frac{1}{x^2}$$

$$y = -\frac{1}{x^2}$$



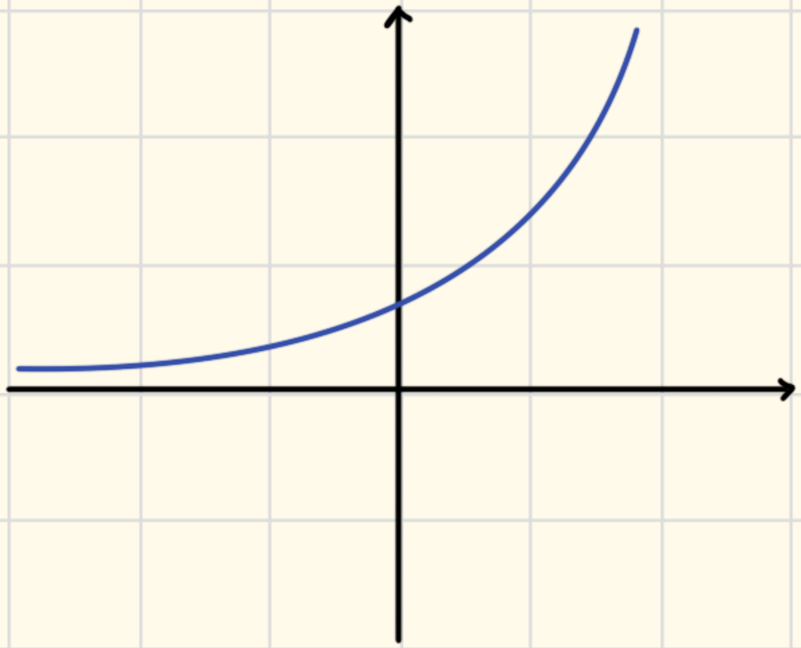


EXPONENTIAL

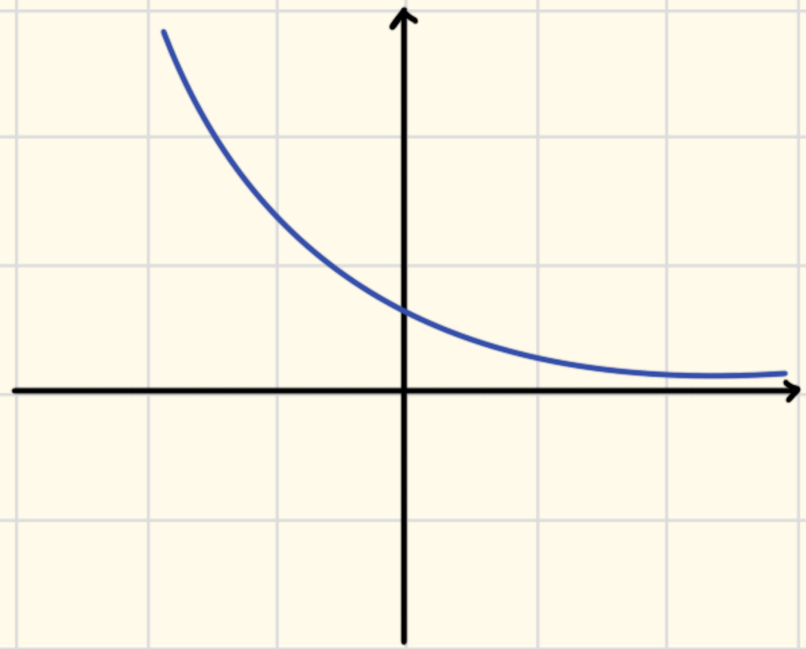
Exponentials model very fast growth OR decay

$$y = a^{x}$$

$$y = a^{-x}$$



Growth



Decay

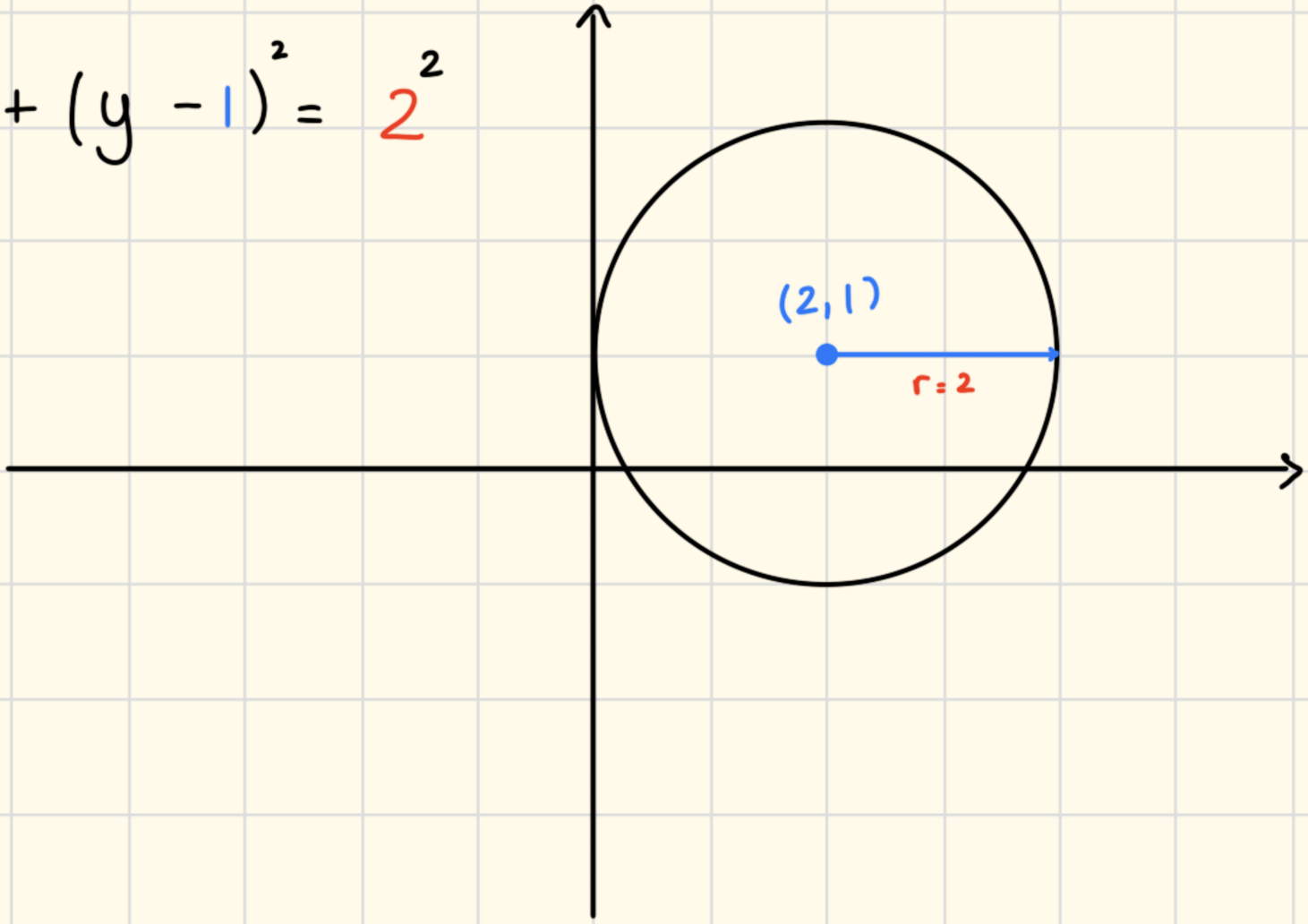
CIRCLE

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

center (x_0, y_0)

radius

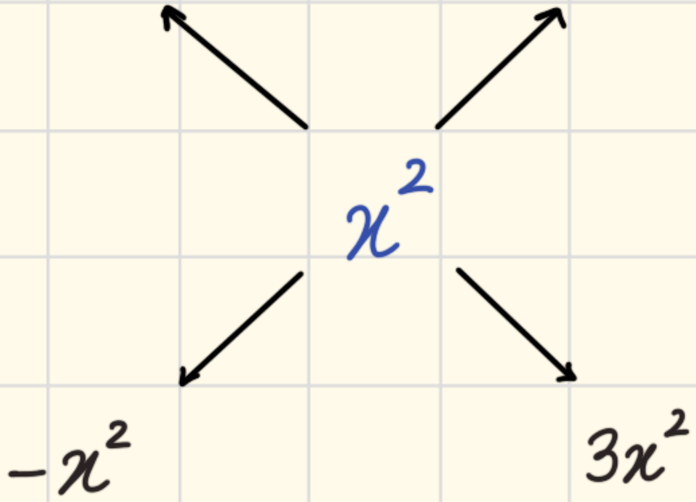
$$(x - 2)^2 + (y - 1)^2 = 2^2$$

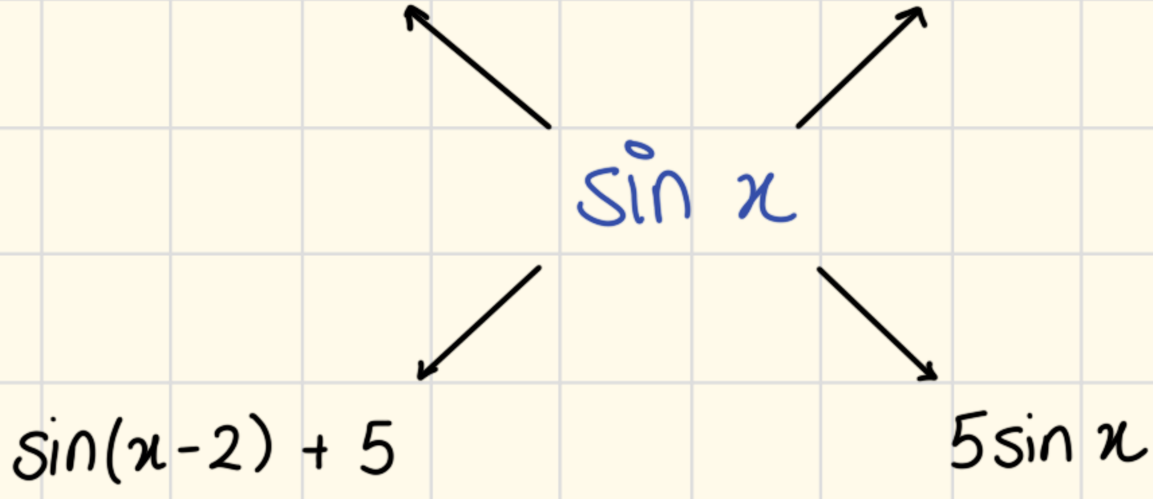


Functions And Graphs

You can build a family of functions from parent functions.

$x^2 + 2$ $(x-1)^2 + 5$ $\sin x - 1$ $\sin(2x)$



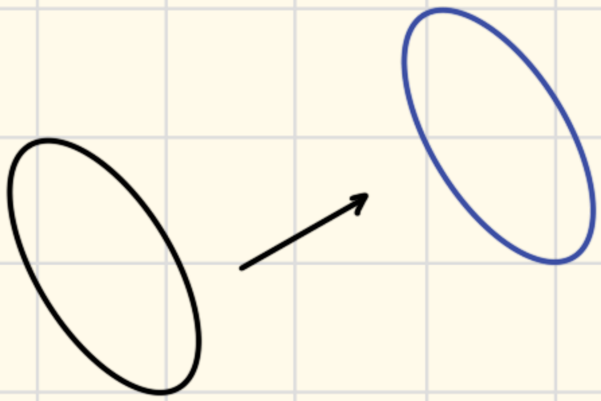


These are called TRANSFORMATIONS!

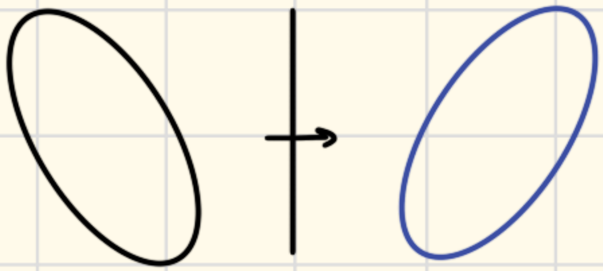
Graph Transformations

There are 3 types of transformations.

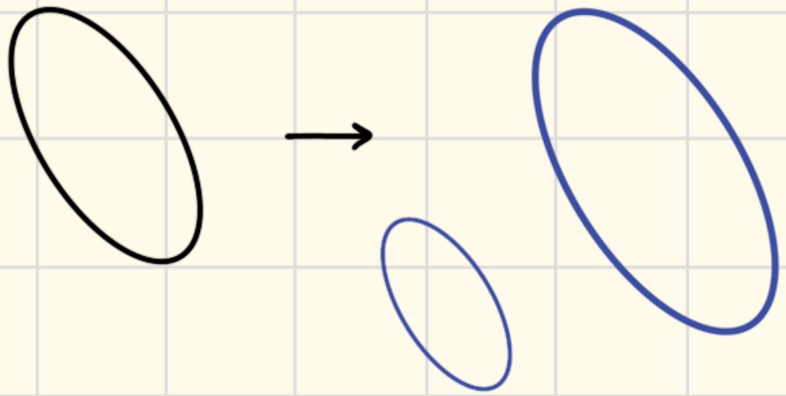
Translation



Reflection



Stretch And Compress



Graph Transformations

We can either transform through the x OR y.

Change in:

x

Opposite of what you expect

y

Exactly what you expect

TRANSLATIONS

Translations slide curves left, right, up and down.

x

- $f(x-a)$

  ↓

  move right a units

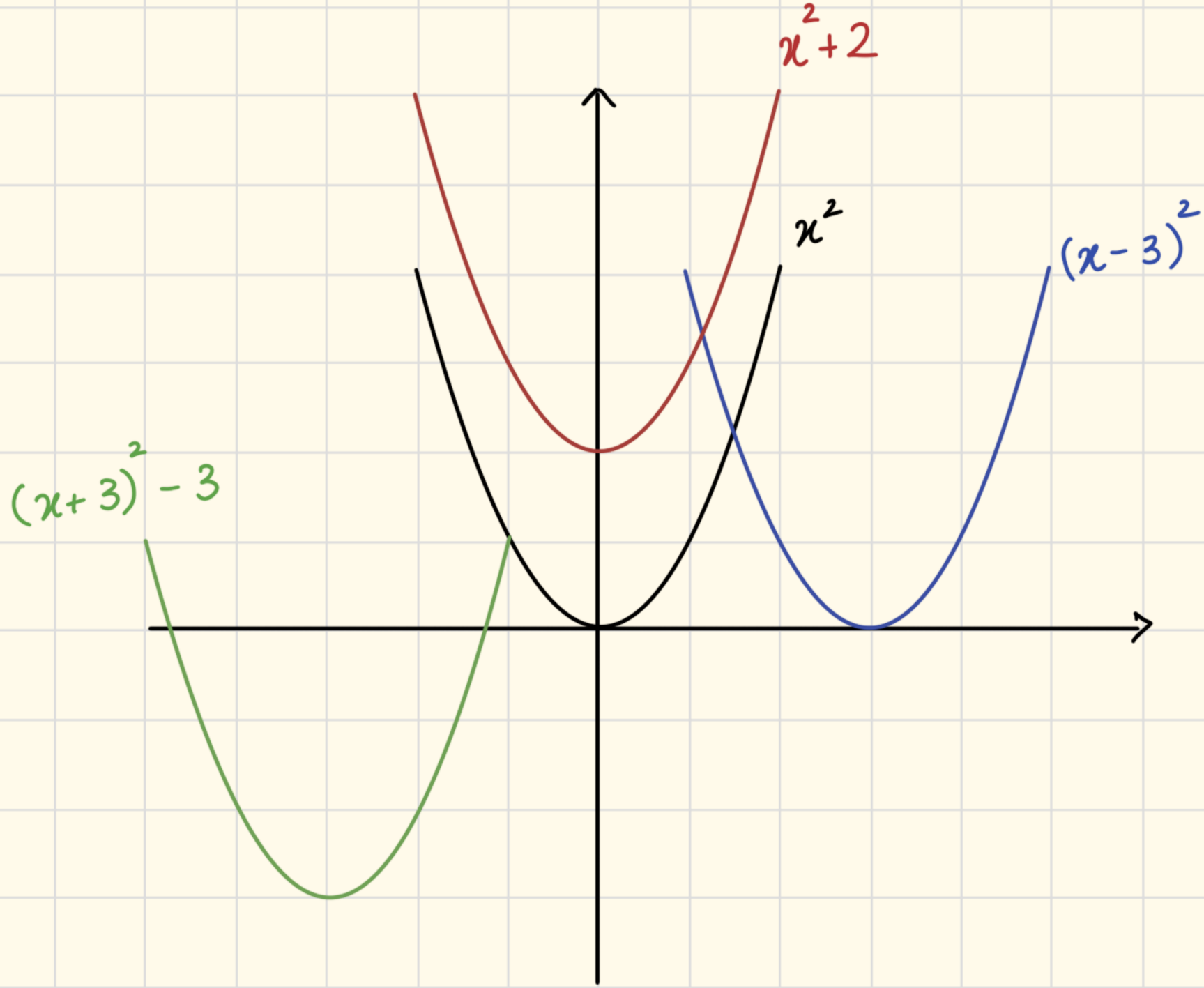
y

- $f(x)+a$

  ↓

  move up a units

TRANSLATIONS



REFLECTIONS

Reflections mirror on the x OR y axis.

x

- $f(-x)$

↓

reflect on y-axis

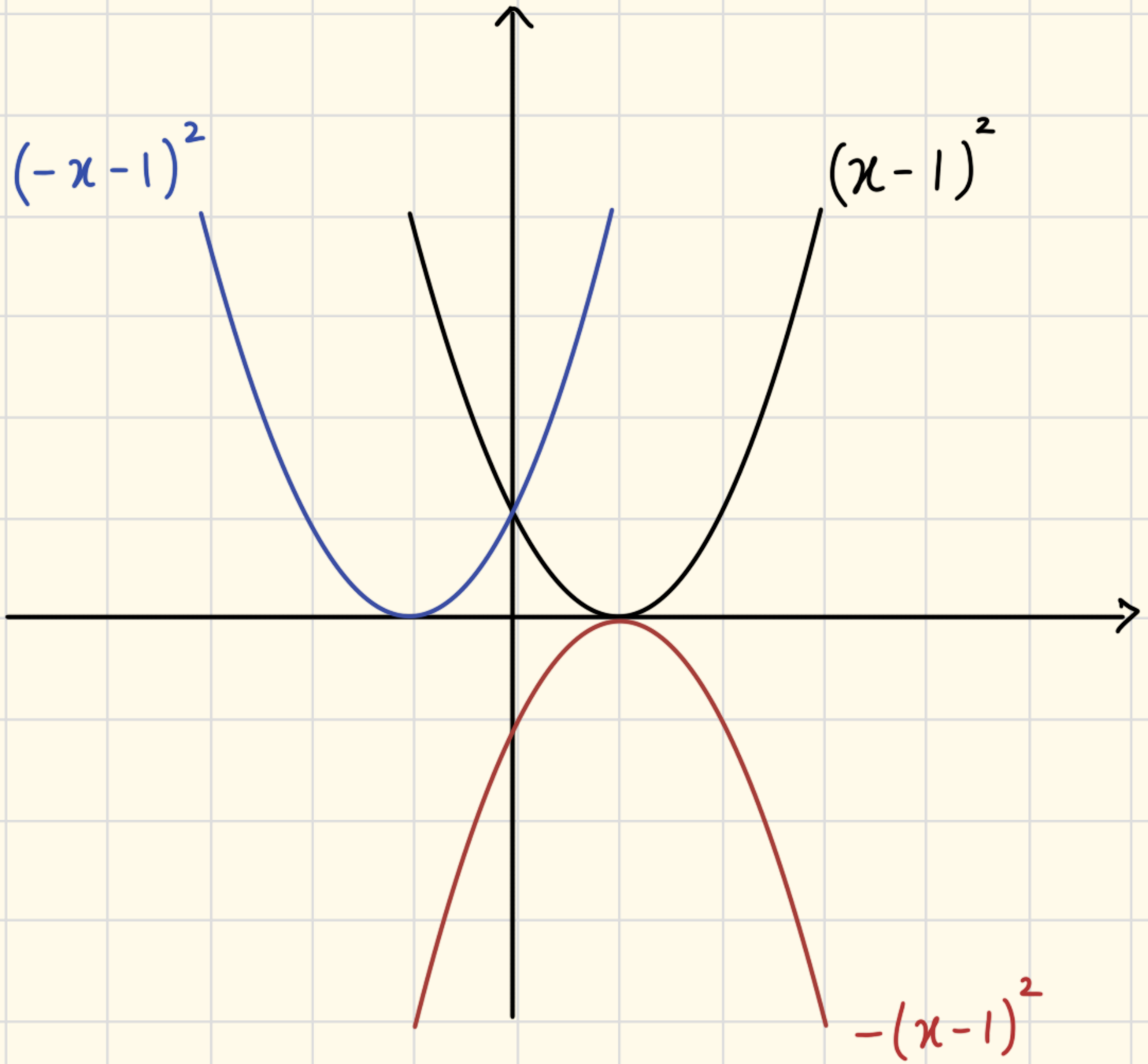
y

- $-f(x)$

↓

reflect on x-axis

REFLECTIONS



Stretch And Compress

Stretch and Compress in the x or y directions.

x y

- $f(ax)$

↓

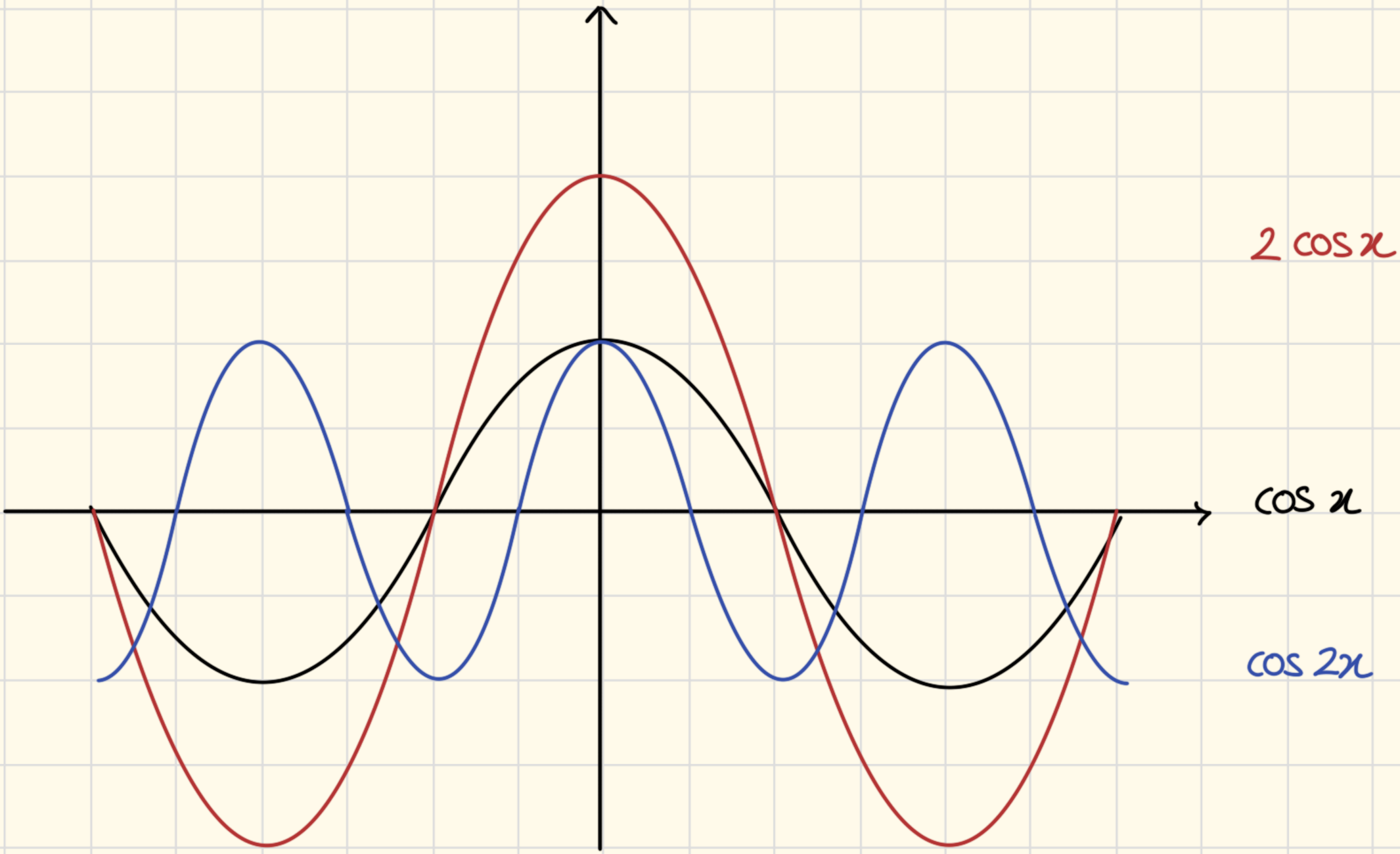
compress in x-direction by factor $\frac{1}{a}$

- $a\,f(x)$

↓

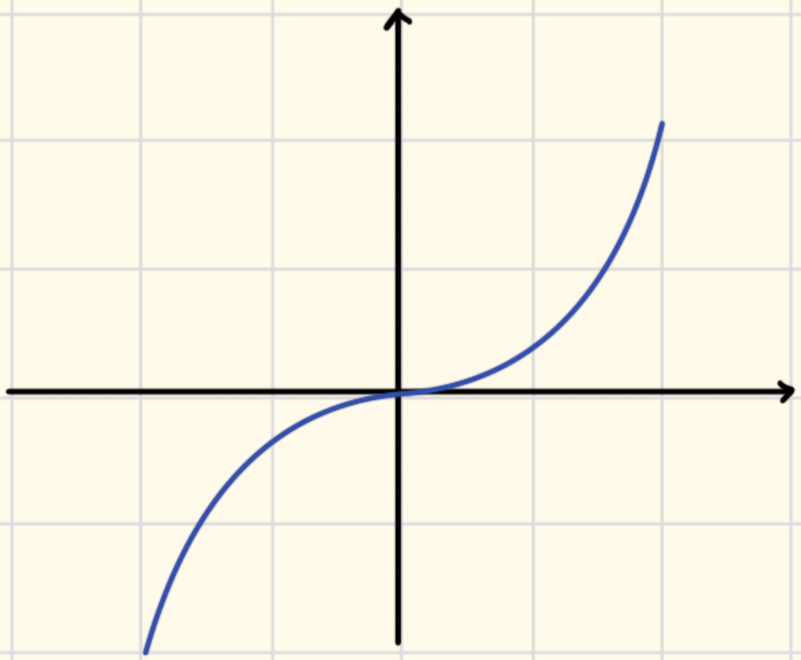
Stretch in y-direction by factor a

Stretch And Compress

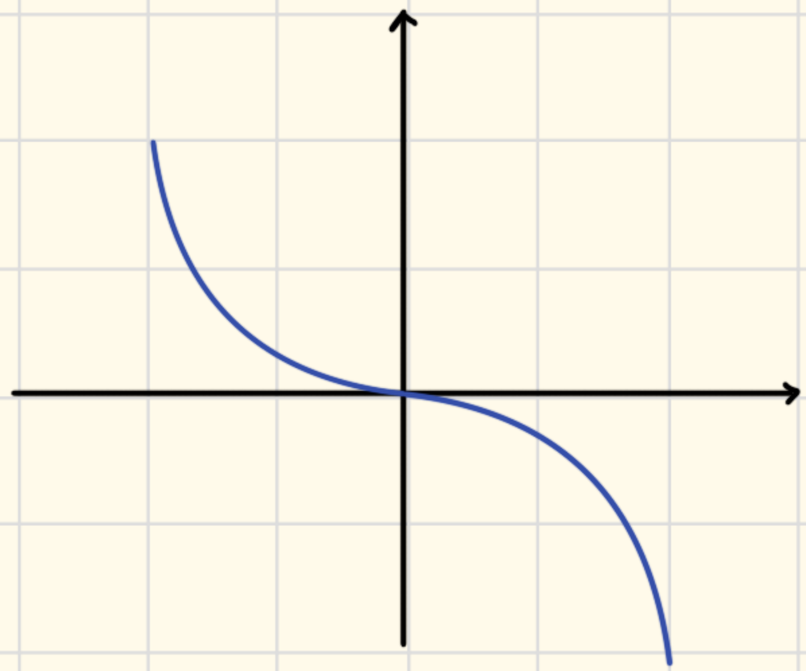


Functions And Graphs

How to know the difference between



x^3 ?



$-x^3$?

Functions And Graphs

You can test a point!

Take $x = 1$,

$$y = x^3 = 1^3$$

positive



$$y = -x^3 = -1^3$$

negative

